ARM-450 — Part-by-Part Geometry Audit

Read from the STEP B-rep, not from the source that made it and not from a mesh.

> ## STATUS: ALL DEFECTS FIXED — 2026-08-21

> 52 features verified present · 0 needing attention. Nothing had been printed when these were found, so every fix went in at the source and every part was regenerated. The sections in this document are re-rendered from the corrected geometry: all nine now read POCKET PRESENT.

> | # | part | defect | fix | | --- | --- | --- | --- | | 1–2 | link halves | no Ø42 bearing pocket | solid boss ring unioned back before counterboring | | 3 | base | no Ø42 J1 seat | cable bore now stops below the seat, leaving a Ø38 shoulder | | 4 | wrist_j6_output | 3 of 4 tool bolts | honest 3-bolt interface at 160/250/340°, all in material | | 5 | wrist_j5_yoke | cheek spacing 38.02 | SPAN derived from BRG_SPACING → 40.00 | | 6 | wrist parts | horn_pattern() no-ops | still open — needs your decision, see §6 |

> Mass went 1143 g → 1191 g. The boss rings cost 48 g. That is 9 g under the 1.2 kg limit — real but very tight, and worth knowing before anything else is added.

> The pre-print gate now runs this audit as check 56, so this class of defect cannot pass it again.

audit_parts.py enumerates every cylindrical face in each STEP file, groups the faces into physical holes, and checks each part against the features it is supposed to have. draw_2d_verify.py then cuts a true section through each bearing interface so every finding can be confirmed by eye.

Why the pre-print gate did not catch any of this

The gate contains this check:

```
check(abs((BRG_OD + BRG_FIT) - 42.00) < 1e-6, "bearing pocket Ø42.00", ...)
```

That checks the parameter. It confirms two numbers in params.py add up to 42.00. It never opens the part. Every one of the defects below passed that gate 55/55, because in each case the parameter is perfectly correct and the geometry does not contain it.

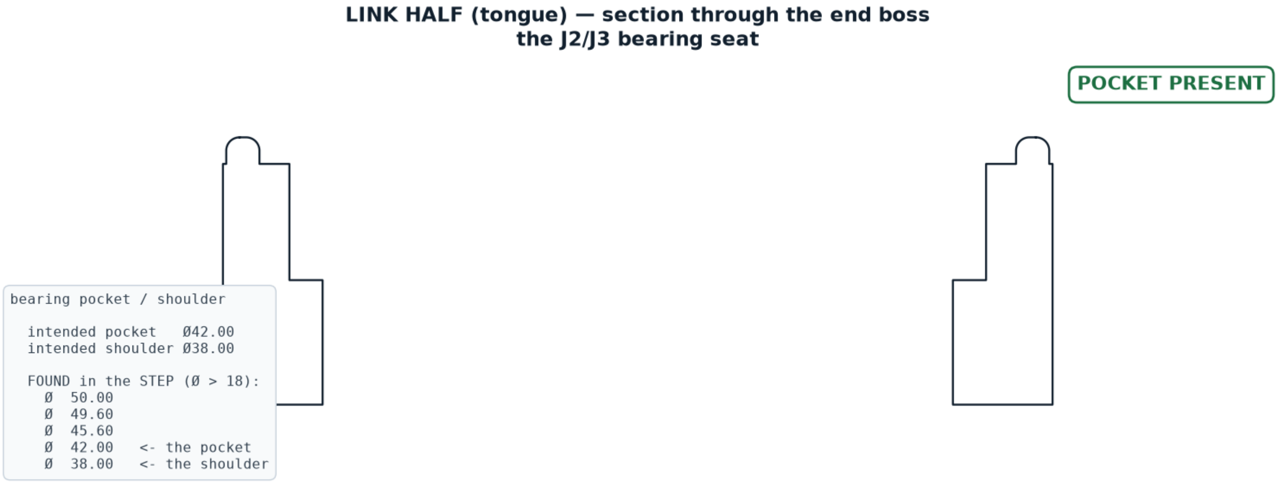
A gate that reads the design intent instead of the produced geometry cannot find a feature that was silently consumed by an earlier cut. That is the single most useful thing this audit established, and it is why audit_parts.py now exists alongside the gate.

What was found, and what it was

#	part	defect	severity
1	link_half_tongue	no Ø42 bearing pocket	BLOCKING
2	link_half_groove	no Ø42 bearing pocket	BLOCKING
3	base	no Ø42 J1 bearing seat	BLOCKING

#	part	defect	severity
4	wrist_j6_output	3 of 4 tool-face bolt holes	moderate
5	wrist_j5_yoke	cheek spacing 38.02, drawing says 40.00	minor
6	all wrist parts	horn_pattern() calls are no-ops	needs a decision

1 & 2 — The link halves have no bearing pocket

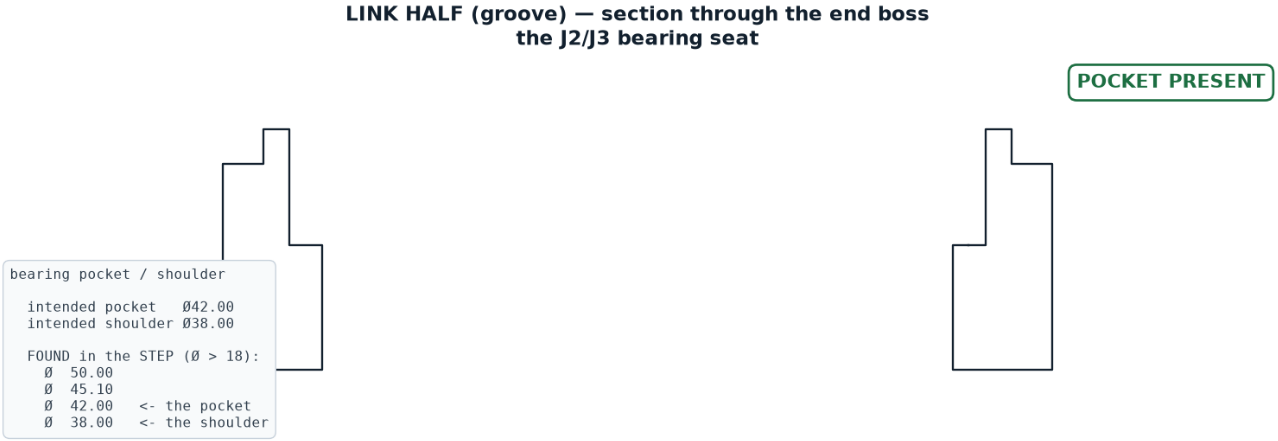


The section is taken straight through the end boss. There is no counterbore step. The wall runs straight from the web to the seam face.

Diameters the STEP actually contains at that boss: Ø50.00, Ø49.60, Ø45.60, Ø45.20, Ø38.00. There is no Ø42.

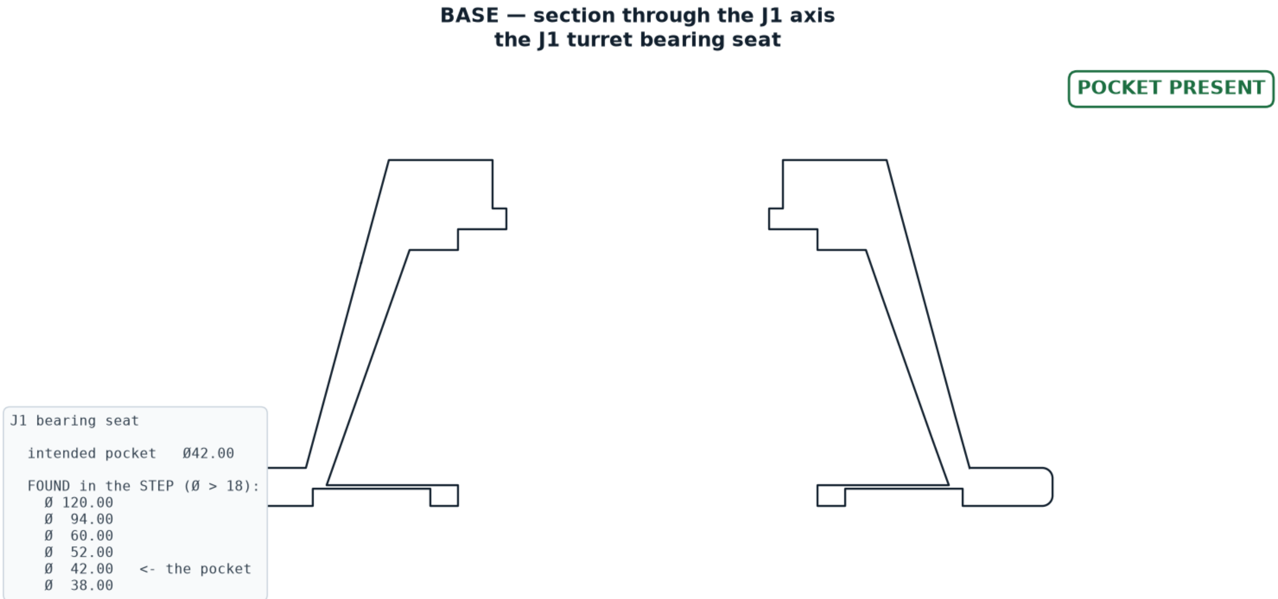
Cause. In cad/link_shell.py the shell is hollowed at step 2 and the pocket is cut at step 3. The hollow uses boss_r - WALL_FLANGE = 25 − 2.4 = Ø45.20 at the end boss. The pocket cut that follows is Ø42.00 — smaller than the hole it is cutting into, so it removes nothing at all. CadQuery does not complain: cutting a small cylinder out of an already-empty region is a legal no-op.

Consequence. A Ø42.00 bearing dropped into a Ø45.20 bore has 1.60 mm of radial clearance. At the 360 mm tool radius that is roughly 20 mm of tool wander — the joint has no location at all. This is the exact failure the paired-bearing redesign was written to eliminate, and it has been present in every version of the link.



Identical on the groove half.

3 — The base has no J1 bearing seat



Both pedestal walls end in a plain flat rim. There is no step for a bearing to sit in. Diameters found: Ø120.00 and Ø52.00. No Ø42.

Cause. In cad/base_wrist.py:

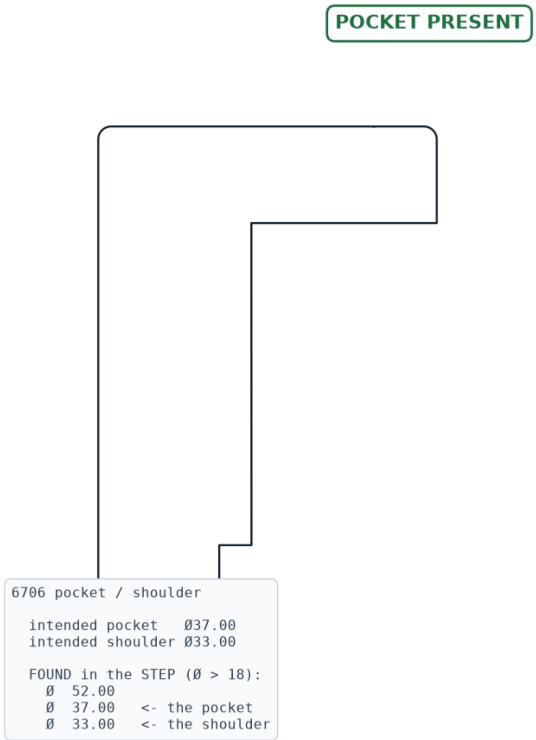
```
b = b.cut(... circle(CABLE_BORE / 2) ...)   # Ø52, straight through
b = b.cut(... circle((BRG_OD + BRG_FIT) / 2) ...)   # Ø42 "seat"
```

The Ø52 cable bore is cut through the full height first. The Ø42 seat is then cut inside a hole that is already Ø52. 42 < 52, so it removes nothing.

Consequence. The J1 bearing has nothing to seat in — a Ø42 bearing over a Ø52 hole drops straight through. The whole arm has no defined rotation axis at its base.

4 — J6 tool face has three bolt holes, not four

J6 OUTPUT — section through the bearing axis



Holes found at r = 15.00 (correct, Ø30 BCD): (−10.61, −10.61), (−10.61, +10.61), (+10.61, −10.61). The fourth, at (+10.61, +10.61), is absent — the servo pocket opening occupies that quadrant, so there is no material to drill.

The isometric sheet calls out "4 × Ø3.40 ON Ø30.00 BCD". Three of four means a tool bolts on asymmetrically, and the drawing overstates what is there.

5 — J5 yoke cheek spacing

Bearing-face spacing measures 38.02 mm. BRG_SPACING is 40.00 and the isometric sheet says "CHEEK SPACING 40.00". Moment stiffness goes as spacing squared, so 38.02 against 40.00 is a 9.6 % loss of moment stiffness — small, but the drawing should not claim a number the part does not have.

6 — The horn patterns are no-ops

horn_pattern() is called on wrist_j4_housing, wrist_j6_output and turret_j1. None of the three parts contains a single hole on the Ø14 horn bolt circle. The pattern is cut at r = 7.0, which in every case falls inside the Ø38 through bore or the Ø42 pocket —

already-empty space, so again a legal no-op.

This one needs a decision rather than a fix: if the servo body is captured in the pocket and its horn bolts to the mating part, no horn holes are needed here and the calls should be deleted. If the horn is meant to drive this part, the holes have to move outboard of the Ø38 bore.

The pattern behind all of it

Every one of these is the same mistake in a different place: a cut made into a region that a previous cut had already emptied. Boolean subtraction of a smaller shape from a larger void is silent — no error, no warning, no change in volume. It looks exactly like success.

Three defences now exist:

1. audit_parts.py — reads geometry, not intent, and checks that each expected feature is physically present at the right size and place.
2. draw_2d_verify.py — a section through every bearing interface, so a missing seat is visible rather than inferred.
3. cad/volumes.py — a single generator for the mass table, so derived numbers cannot go stale unnoticed.

All of findings 1–5 are now fixed and verified. In each case the fix was ordering: restore the material before counterboring it, or stop the earlier cut short of the feature it would otherwise swallow.

Finding 6 is a design decision and is still open — see above.

The durable change is check 56 in preprint_check.py: the gate now runs this audit and fails on any missing feature. Checking the parameter was never going to be enough.